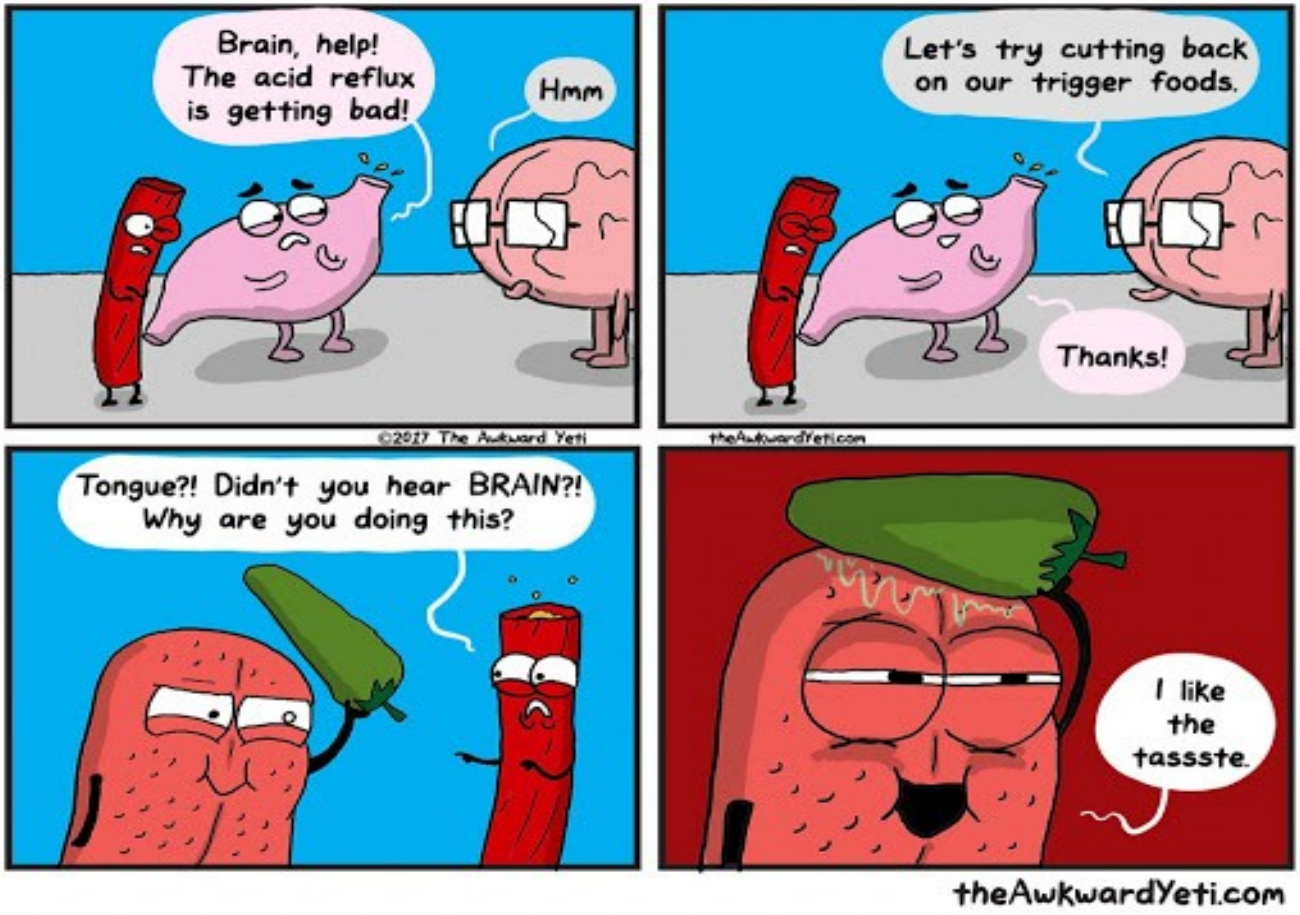
DRUGS USED IN ACID PEPTIC DISEASES

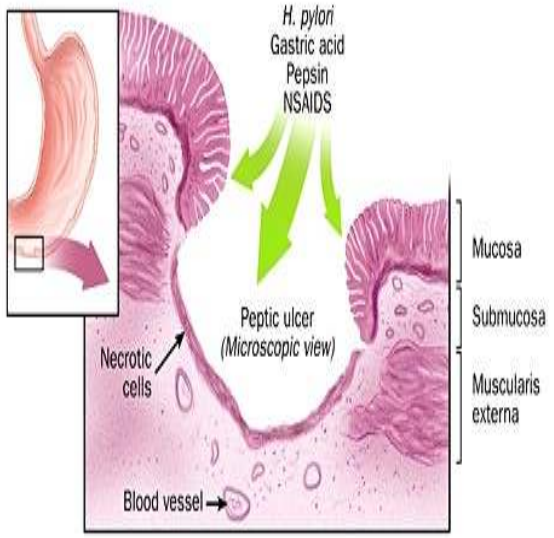


Learning Objectives

- **Remember:** List the causes of gastric H^+ hypersecretion and diagnostic tests for *H. pylori*.
- **Understand:** Explain H^+ secretion by parietal cells and the role of *H. pylori* in peptic ulcer disease.
- **Apply:** Apply the mechanisms of PPIs and H_2-receptor antagonists to clinical treatment.
- **Analyze:** Compare the mechanisms and selectivity of PPIs and H_2-receptor antagonists.
- **Evaluate:** Assess the appropriate therapeutic uses of PPIs and H_2-receptor antagonists.
- **Create:** Develop an appropriate treatment approach for acid-peptic disorders, including *H. pylori* infection.

PEPTIC ULCER

- acid peptic injury of the digestive tract, resulting in mucosal break reaching the submucosa
- Includes both gastric and duodenal ulcer
- Duodenal ulcer is five times more common than gastric ulcer



	Gastric Ulcer	Duodenal Ulcer
Age	middle age 50-60	Any age (specially 30-40)
Pain	Epigastric, can radiate to back	Epigastric
Onset	Immediately after eating	2-3 hours after eating & at midnight
Agg.by	Eating	Hunger
Relived by	Lying down or vomiting	Eating
Duration	Few weeks	1-2 months
Weight	Wt. loss	No wt. loss

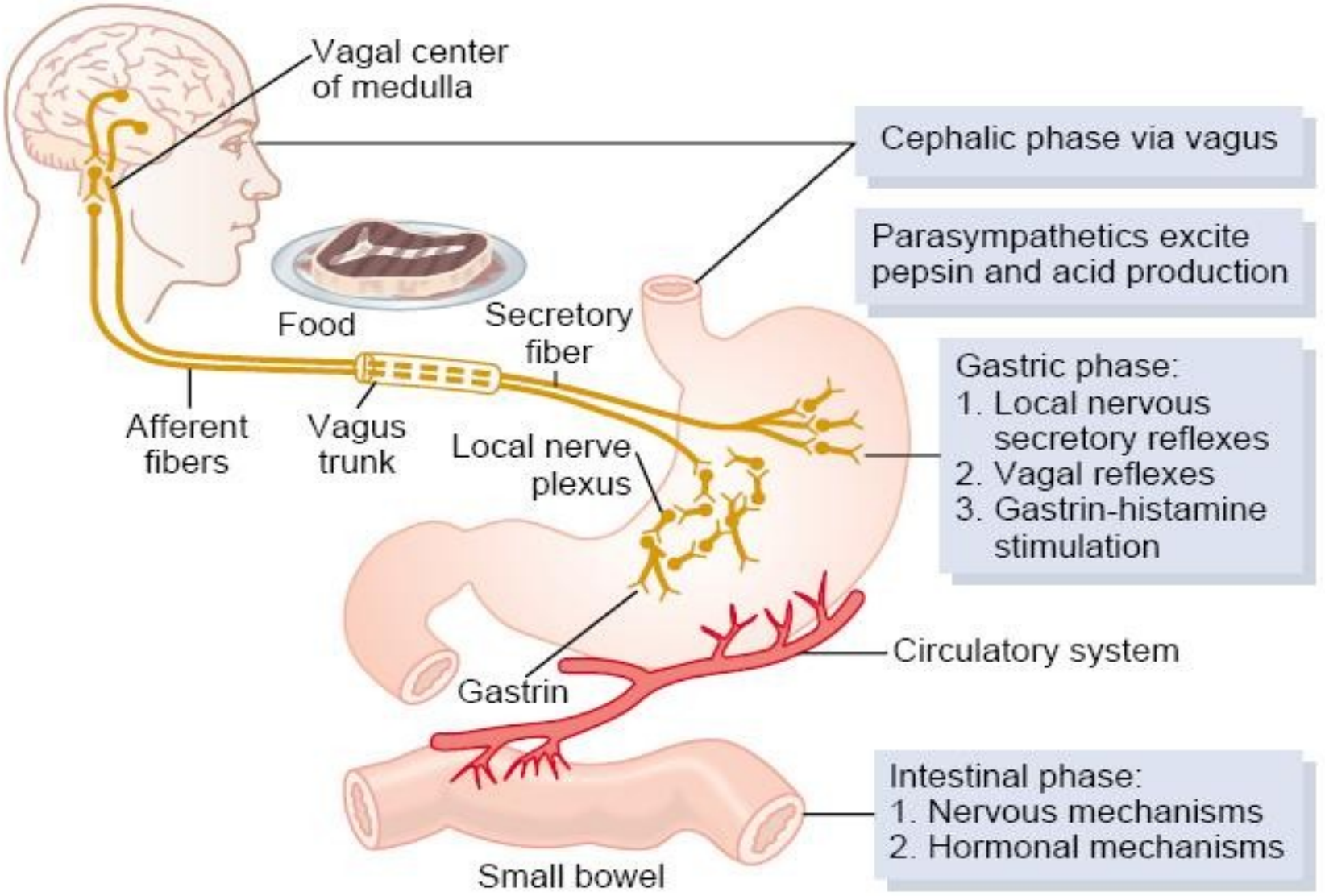
Gastric Secretion



- 2.5 litres of gastric juice daily
- Activates pepsinogen to active enzyme pepsin
- Kills microorganism ingested with food

- Chief cells –pepsinogen
- Parietal cells- secretes HCl and IF
- Mucus neck cells- secretes alkaline mucus and bicarbonate ions

Phases of gastric secretion



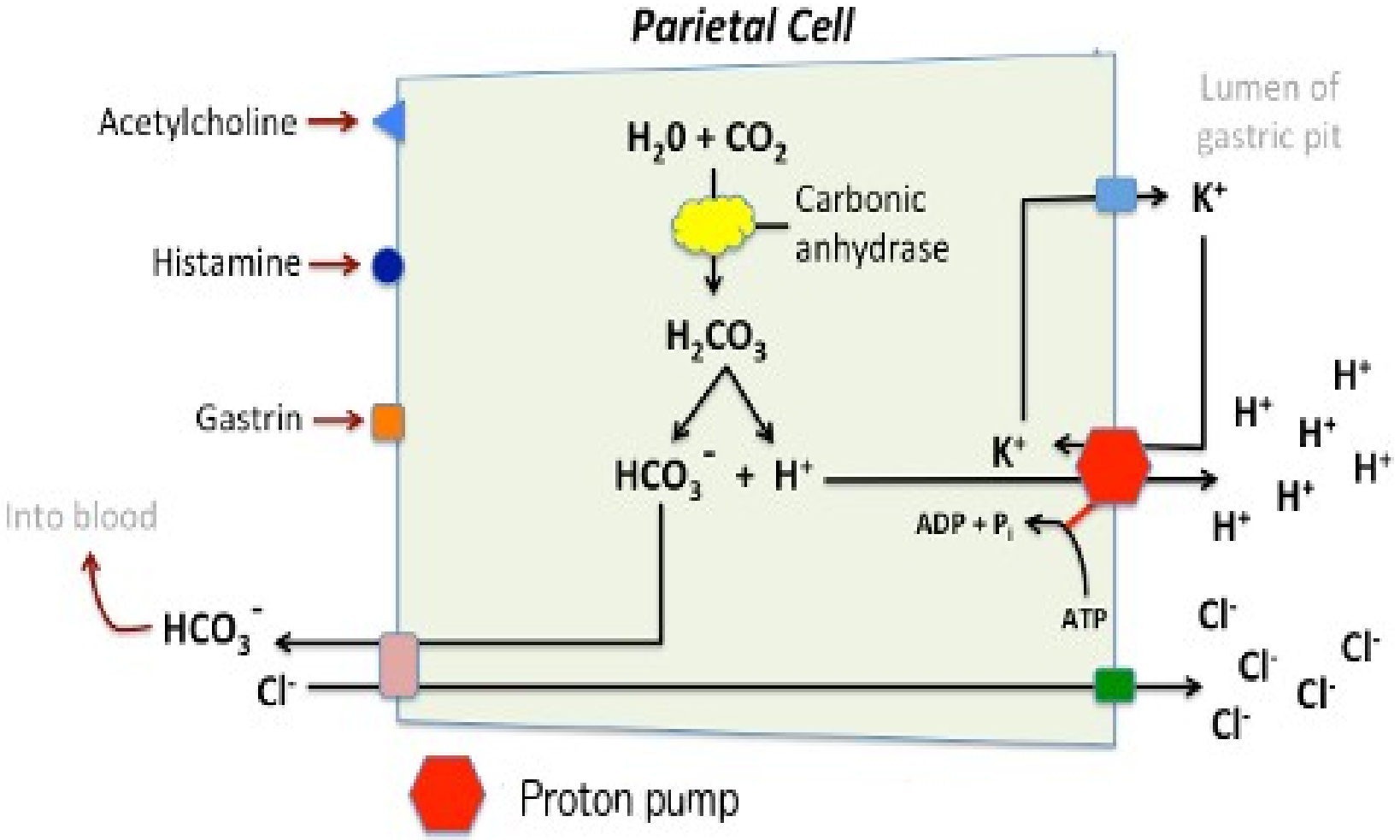
REGULATION OF GASTRIC ACID SECRETION

- The secretion of gastric acid by parietal cells is regulated by **acetylcholine, histamine, gastrin, somatostation and prostaglandin E_2**

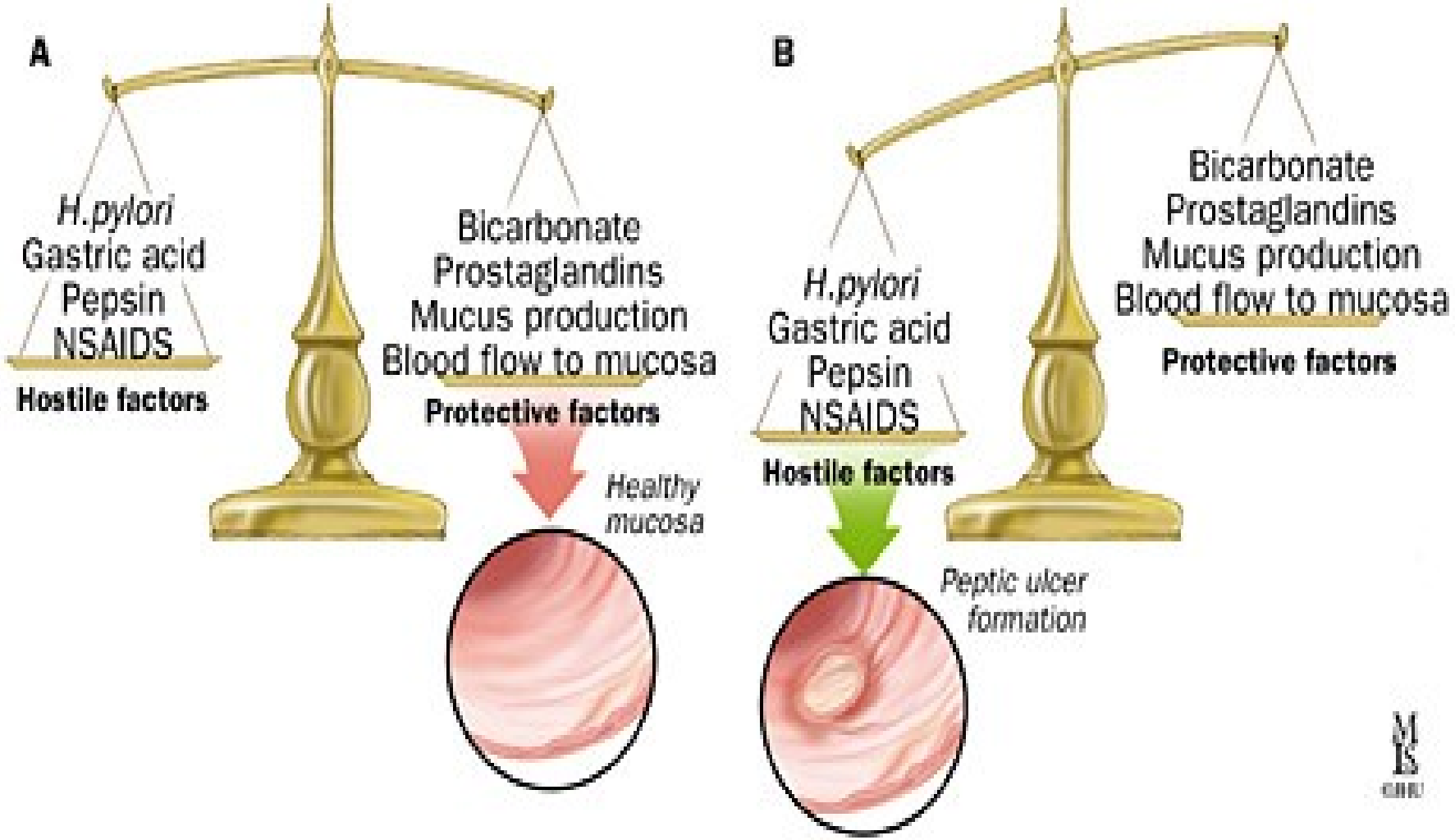
- Binding of **histamine, Ach and gastrin** to their specific receptors on the parietal cell results in the increased secretion of gastric acid

- **Somatostatin** (SST)-antral D cells- Inhibits gastric acid secretion
- Binding of **PGE_2** to its receptor decreases the gastric acid secretion

PHYSIOLOGY OF GASTRIC ACID SECRETION



Pathophysiology of PUD



Helicobacter pylori

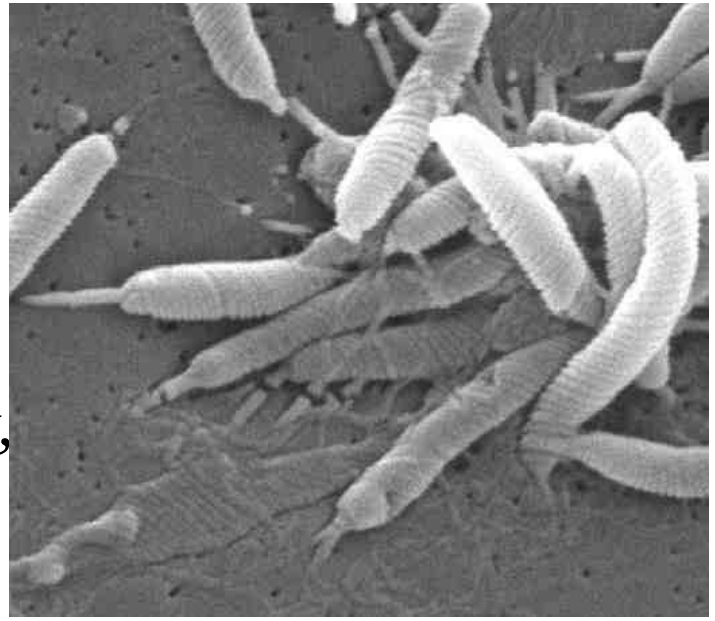
- *H. pylori* accounts for

the majority of PUD



- Gram-negative, spiral shaped, flagellated, microaerophilic rod
- more common in the developing world
- These organism **produces urease to create an alkaline environmen**t, which is essential for it's survival in the stomach under the mucosal barrier.
- **Expresses adhesins** (Blood group antigen adhesin and outer inflammatory protein adhesin)- facilitates the attachment of bacteria to gastric epithelium

Risk factors

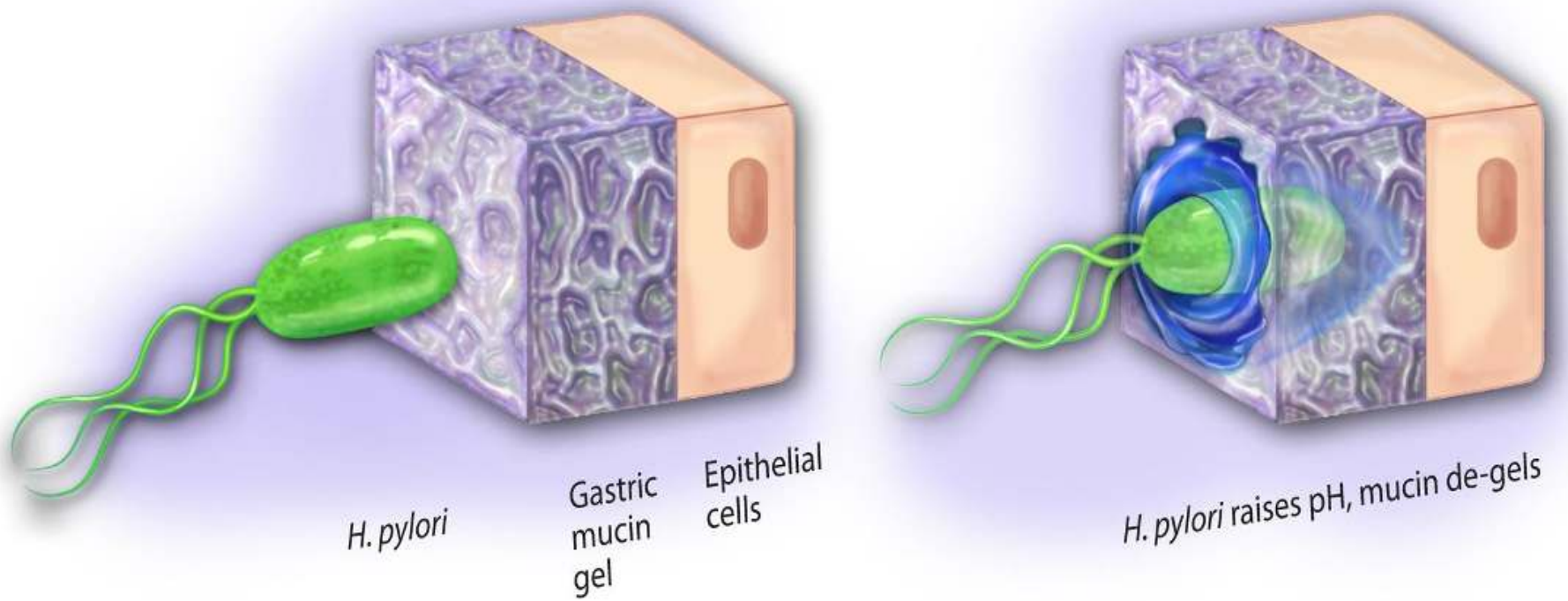


Risk factors for *H. pylori* infection

- poor SES/ less education
- birth or residence in a developing country,
- domestic crowding,
- unsanitary living conditions,
- unclean food or water
- exposure to gastric contents of an infected individual.

Transmission : oral-oral / fecal-oral route

H. PYLORI CROSSING MUCUS LAYER OF STOMACH



NSAID-Induced ulcer



- Most commonly used medications
- Major risk factors liked to peptic ulcer disease and its complication
- Increases the risk by four times in case of NSAID users and two times in case of aspirin users
- Concomitant use of NSAIDs or Aspirin with other drugs (SSRIs, corticosteroids, anticoagulant) increases the risk of upper GI bleeding

- COX-1 is important in the **production of prostaglandins.**
- Prostaglandins have a number of protective effects on the gastric mucosa such as the inhibition of enterochromaffin-like cells (ECL).
- This inhibition reduces the secretion of histamine that stimulates parietal cells to secrete hydrochloric acid (HCl).
- NSAIDS causes **inhibition of COX-1** reduces the production of mucosal protective prostaglandins- more secretion of HCl
- Common cause of GI disturbance

Other causes…





- Cigarette smoking
- Genetic predisposition
- Diet
- Ischaemia, drugs, radiotherapy, Zollinger Ellison syndrome

Clinical features

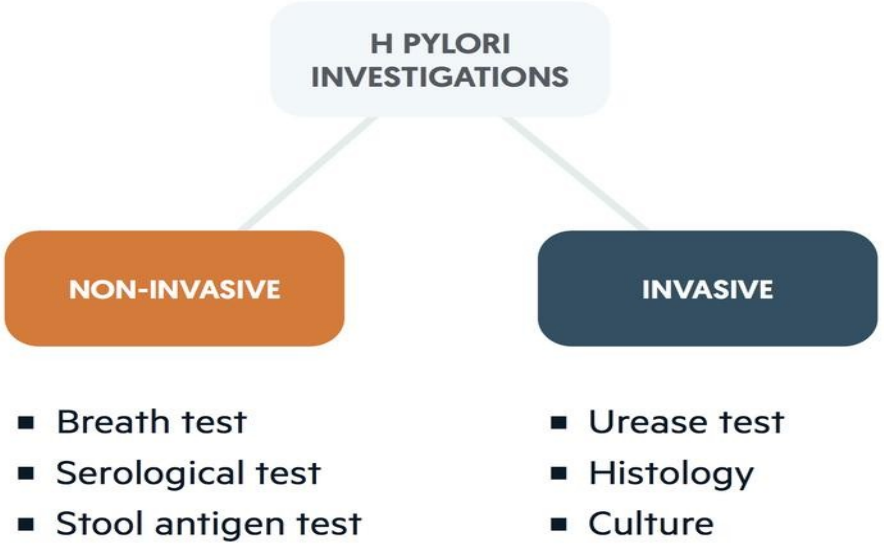
- Nocturnal abdominal pain(duodenal ulcer)
- Epigastric pain, nausea, vomiting, weight loss(gastric ulcer)

Main complication of PUD

- Bleeding- melena or haematemesis
- Perforation- sudden onset of intense pain in upper abdomen
- Gastric outlet obstruction- nausea and vomiting

Diagnosis

- Endoscopy of the upper GI- gold standard for diagnosis
- Detection of ***H.Pylori*** infection- essential for treatment plan



Management

- The old dictum “no acid- no ulcer”
- Although acid secretion is still important in the pathogenesis of PUD,

-Eradication of *H. pylori* and treatment of NSAID-induced ulcer-is the mainstay of treatment

CLASSIFICATION

I. Drugs which neutralise gastric acid (Antacids)

a) Systemic antacids: Sodium bicarbonate

b) Non systemic antacids:

i. Buffer type: Aluminium hydroxide, Magnesium trisilicate, magaldrate

ii. Non buffer type: Magnesium Hydroxide, Calcium carbonate

iii. Miscellaneous: Alginates, Simethicone

II. Drugs which reduce the gastric acid secretion

a) H2 receptor antagonists

b) Proton pump inhibitors

c) Prostaglandin analogues

d) Anticholinergics

III. **Mucosal protective agents**: Sucralfate, Colloidal bismuth subcitrate

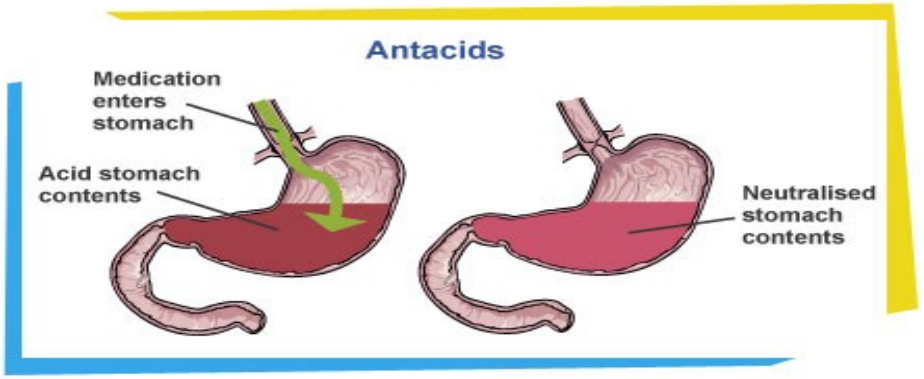
III. **Ulcer healing drug:** Carbenoxolone sodium

III. **Anti H.pylori drugs**

Presently ,Proton pump inhibitors and H2 receptor blockers are most widely used….

ANTACIDS

- Weak bases-neutralise gastric acid and raise the pH of gastric contents
- Also inhibits formation of pepsin (pepsinogen is converted to pepsin at acidic pH)
- Stimulates production of PGs
- Given between meals and at bed time
- Divided into 2 classes: systemic and non systemic



TYPES

Sodium bicarbonate :

- Rapidly acting
- Raises pH-7
- Not indicated for long term use because:

- It releases CO2 (causes gastric distention ,belching,)
- It forms NaCl- increases the sodium load- fluid retention

- Absorbed in systemic circulation- systemic alkalosis
- Rebound acidity can occur

Non systemic antacids:

- Poorly absorbed from GIT
- Slow onset but longer duration
- Rebound acidity does not occur



COMBINATION OF ANTACIDS

- Aluminium hydroxide and magnesium hydroxide

1. Magnesium hydroxide- fast acting
   Aluminium hydroxide- slow acting
Combination provides prompt as well as sustained effect

2. Magnesium hydroxide- diarrhea
Aluminium hydroxide- constipation
Combination (bowel movement is least affected

3. Dose of individual component is reduced
Systemic toxicity is minimised



PROTON PUMP INHIBITORS

Omeprazole, Esomeprazole, Lansoprazole,Rabeprazole Pantoprazole

- Omeprazole- prototype
- Prodrugs- administer orally as sustained release
- Potent suppressors of gastric acid secretion
- Diminish daily acid production by 80-85%.
- Acid labile- destroyed by gastric acid- available in enteric coated formulation



Pharmacokinetics

- Given orally, pantoprazole can be given I.V
- Food decreases bioavailability- hence given in empty stomach
- Short half life- 1-2 hrs but action lasts upto 24 hours
- Undergoes rapid first pass metabolism
- Metabolised by liver

MECHANISM OF ACTION

Absorption into systemic circulation

Prodrug diffuses into parietal cells canaliculi

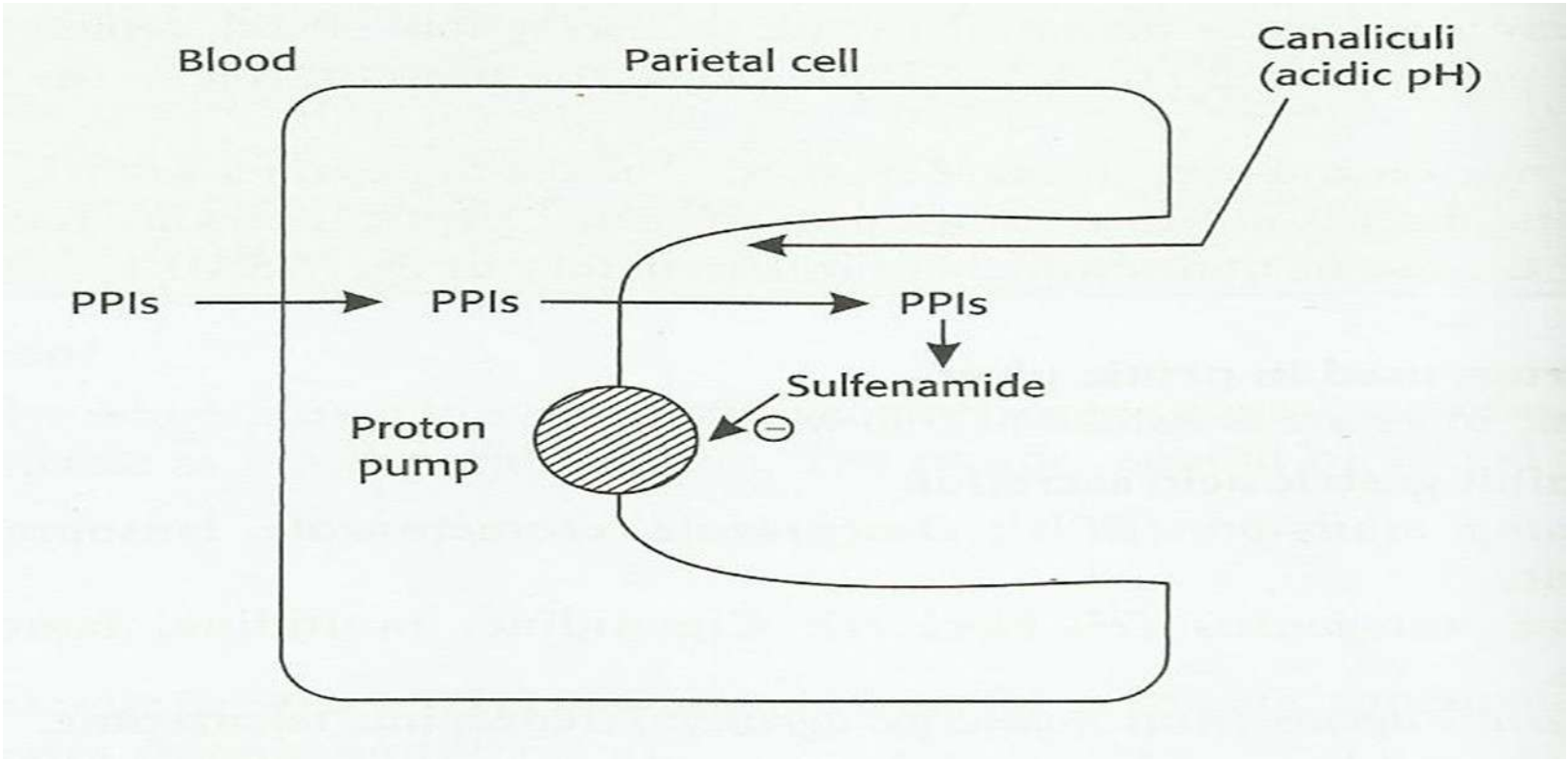
Converted into an active sulfenamide molecule- drug trapping

Contd ...

Irreversible inactivation of proton pump producing acid suppression

Acid secretion resumes - after new pump molecules are synthesized

Prolonged suppression of acid secretion (24-48 hrs) though short plasma t1/2 (0.5-2 hrs)


Blood
Parietal cell
Canaliculi
(acidic pH)
PPIs
PPIs
PPIs
Sulfenamide
Proton
pump

Dosage

Features	Omeprazole	Esomeprazole	Lansoprazole	Pantoprazole	Rabeprazole
Bioavailability	55	70	85	77	52
DOSE					
Duodenal/ gastric ulcer	**20 mg OD** 40 mg OD	**20 mg OD** 20 mg OD	15 mg OD 30 mg OD	**40 mg OD** 40 mg OD	20 mg OD 20 mg OD

Uses

- **Peptic ulcer –**Duodenal and gastric ulcer
- **Gastroesophageal reflux disease –** More effective then H_2 blockers
- **NSAIDs induced ulcer**
- **Zollinger –Ellison syndrome –** Higher dose is required for healing ulcer & more effective then H_2 blockers
- **H. pylori associated ulcer-** combination with antibiotics for heal ulcer & eradicate organism

Adverse effects

- Few- diarrhoea, headache and abdominal pain
- Long term omeprazole therapy-

 Decreased absorption of Vit B12

 Hip fractures(reduced Ca++ absorption)

- Drug interactions

 Decrease in gastric acidity due to PPIs- affects bioavailability of drugs(ketoconazole and digoxin)

 Omeprazole inhibits CYP2C19 and CYP3A4- decrease the metabolsim of warfarin, diazepam , phenytoin

H2 antagonists

Cimetidine, Ranitidine, Famotidine, Nizatidine, Roxatidine

- Cimetidine- prototype
- They block > 90% nocturnal acid secretion and 60-70% day time food stimulated acid secretion
- Less potent than PPI’s
- Famotidine- most potent H2 blocker
- Nizatidine- 100% bioavailability



Cimetidine- not used routinely because:

a. It can cross BBB- results in mental state change
b. Inhibits binding of dihydrotestosterone to androgen receptors- impotence in males
c. Inhibits metabolism of estradiol- increases serum prolactin levels- gynaecomastia(males) and galactorrhoea(females)
d. Potent inhibitor of CYP enzymes- increases the plasma concentration of warfarin, theophylline, etc
e. Least potent H2 blocker

- Ranitidine:
  - About 5 times potent than cimetidine
  - No anti androgenic actions
  - Lesser permeability to brain
  - Lower overall side effects
- Famotidine:
  - 5-8 times more potent than ranitidine



MECHANISM OF ACTION

Reversibly competes with histamine

Binds to and inhibits H2 receptors on basolateral membrane of parietal cell

Inhibits acid secretion

Uses of H2 blockers

- **Duodenal ulcer (DU)**

  60-85% of ulcer heals in 4 weeks of therapy

  70-95% of ulcer heals in 8 weeks of therapy
- **Gastric ulcer(GU) –** healing rate some what lower , 50-75% at 8 weeks
- **Stress ulcer & gastritis**
- **Zollinger –Ellison syndrome –** Proton pump inhibitors(PPIs) are DOC
- **Gastric esophageal reflux disease –** less effective then PPIs.

Features	Cimetidine	Ranitidine	Famotidine
Relative potency	1	5	30
Bioavailability	80	50	50
CYP P450 inhibition	++++++	+	
Dose			
Duodenal /gastric ulcer	800 mg HS/ 400 mg BD	300 mg HS/ 150 mg BD	40 mg HS/ 20mg BD

Adverse effects

- Headache, fatigue, myalgia, constipation
- Cimetidine- displace dihydrotestosterone from receptor and increases plasma prolactin- gynaecomastia in males and galactorrhoea in females

Drug Interactions:

- Antacids- reduces the absorption of all H2 blockers- gap of atleast 2 hrs should be kept
- Cimetidine- potent inhibitor of CYP1A2, CYP2C9, CYP2D6, CYP3A4- plasma half lives of warfarin, theophylline, phenytoin ,etc may be prolonged

Prostaglandin analogues

- PGE2 and PGI2
- Acts as anti-ulcer by:

- Inhibits H+K+ATPase and decrease the acid secretion
- Increasing the release of mucus and bicarbonate
- Increasing the mucosal blood flow



Misoprostol

- PGE1 analogue
- Most specific drug for the treatment and prevention of NSAID induced ulcer.

Enprostil, Rioprostil

PGE2 analogue

Adverse effects-

- Diarrhea and colicky abdominal pain(increases water and electrolyte secretion in intestine)
- Contraindicated in pregnant woman- causes uterine contraction

Anticholinergics

- **Pirenzipine, Telenzipine**
- Selective M1 receptor blocker
- Act by blocking M1 receptor at paracrine cells in gastric mucosa
- Heal and prevent the recurrence of duodenal ulcer
- Anticholinergic side effects- very few
- Telenzepine is 25 times more potent than pirenzipine.

Ulcer protectives-Sucralfate



- It is an aluminium salt of sulfated sucrose
- Polymerizes at pH <4, by cross linking of molecule form sticky gel like consistency- adheres to base of ulcer- acts as physical barrier- prevents the acid exposure
- Not given with antacids- antacids raises the pH by neutralizing the gastric acid
- Not effective in NSAIDs induced ulcer
- Side effects- Constipation, Hypophosphataemia, dry mouth
- It inhibits the absorption of tetracyclines, FQs, H2 blockers, phenytoin
- Other uses- gel form in case of burn dressing and bed sores

Colloidal Bismuth Subcitrate

- Useful in the management of peptic ulcer because:
- Forms acid resistant protective coat over the ulcer base
- Promotes the secretion of protective factors like PGE2, mucus, bicarbonate secretion- neutralizes the acid and reduces the severity of the injury
- Dislodges H.pylori from surface of gastric mucosa and kills it
- 120 mg orally, 4 times daily
- Anti H.pylori treatment
- Excellent safety profile; blackening of stools
- Antacids- decreases it's efficacy



Anti- *Helicobacter Pylori* Drugs

- ***H.Pylori*** – the major risk factor for peptic ulcer
- **Two therapeutic goals for *H.Pylori* associated ulcers**:

 To heal the ulcer

 To eradicate the organism

Combination regimens available:

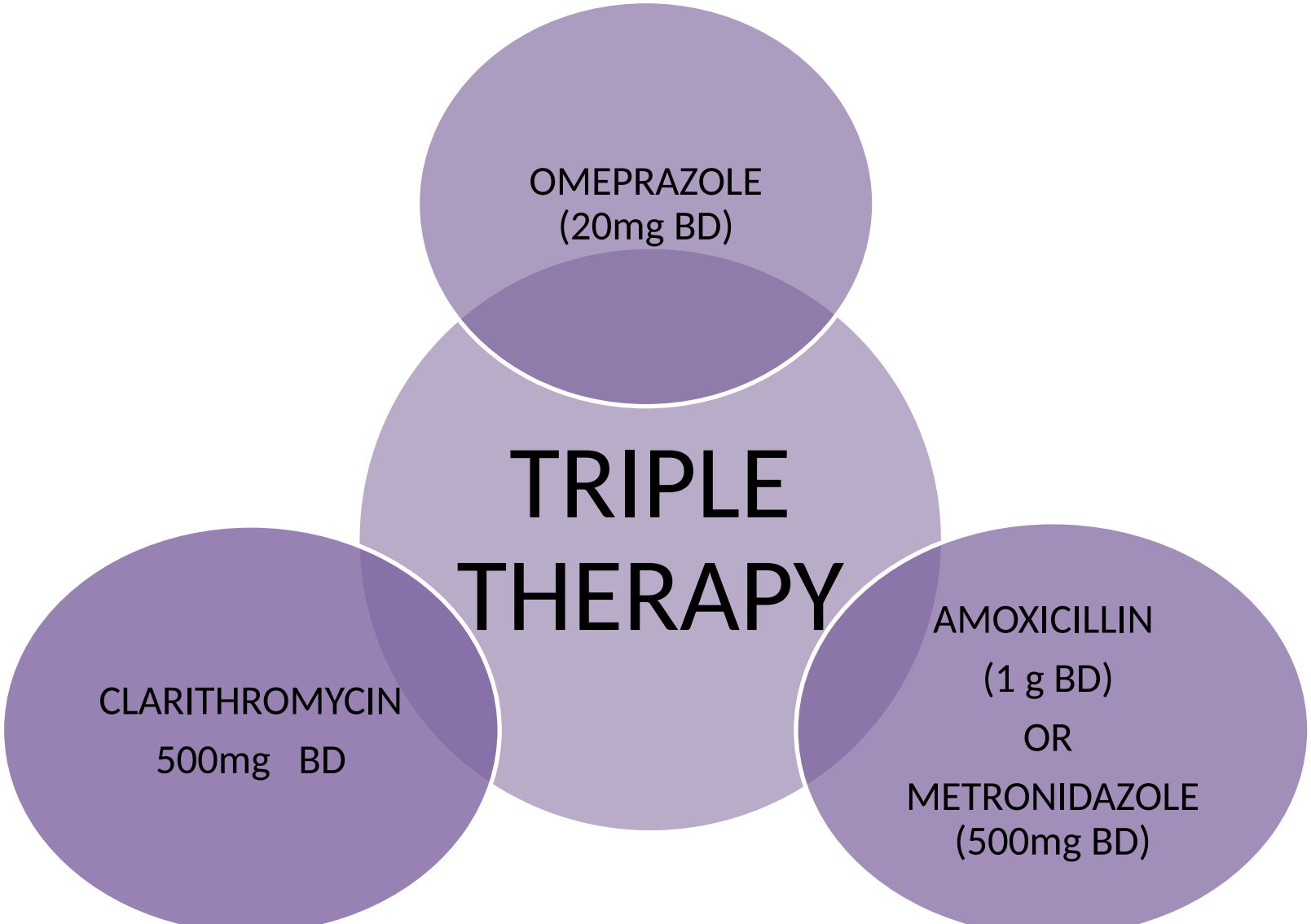
- Triple therapy
- Quadruple therapy
- Sequential therapy

90% eradication rates

- **Usually combination therapy is preferred over monotherpay because:**

- Use of single antimicrobial regimen is ineffective in eradicating the organism and also leads to development of drug resistance
- Combination therapy is associated with better rate of eradication and fast ulcer healing
- Combination therapy is associated with lower relapse rate

Triple therapy- 14 days



Quadruple Therapy -14 days

Omeprazole (20 mg OD)

Bismuth subsalicylate 2 tabs QID (525 mg each)

Metronidazole (250 mg QID)

Tetracycline (500 mg QID)

Sequential therapy- 10 days

Days 1-5

- Omeprazole (20 mg BD)
- Amoxicillin (1 g BD)

Days 6-10

- Omeprazole (20 mg BD)
- Clarithromycin (500 mg BD)
- Tinidazole (500 mg BD)

NSAIDs INDUCED ULCER

- Important cause of peptic ulcer in high income countries where prevalence of *H.Pylori* is declining
- Treatment
- Discontinuation of NSAIDs and use of PPIs for 6-8 weeks
- Those who are using NSAIDS:
- Co-therpy of NSAIDs with PPI, H2 blockers or misoprostol
- Substitution of non-selective NSAIDs with COX-2 selective NSAIDs

* COX-2- selective NSAIDs with PPI- best protection against peptic ulcer complication

Summary

- **Proton pump is stimulated by:**

 Histamine

 acetylcholine

 Gastrin

- **Aluminium hydroxide** causes constipation whereas **magnesium salts** causes diarrhoea
- Long term use of PPIs increases risk of **vit B12 deficiency** and **hip fractures**
- PPIs acts by irreversibly inhibiting the H+K+ATPase in gastric parietal cells
- PPIs have short half lives but can inhibit secretion >24hrs

H2 receptor blockers

- competitively inhibits H2 receptors in the parietal cells and inhibits the acid secretion
- They are more effective in reducing basal acid secretion than stimulated acid secretion
- Famotidine is the most potent while cimetidine is the least potent H2 blocker
- Cimetidine causes gynaecomastia(males) and galactorrhoea(females)

- **Misoprostol(PGE1 analogue)** is the most specific drug for treatment and prevention of **NSAIDs induced ulcer**(but **PPIs is the drug of choice**)
- **Sucralfate should not be given with antacids-** as it polymerises only in acidic medium

- **Triple drug therapy for H.Pylori infection**

 PPI
 Clarithromycin
 Amoxycilin/ metronidazole

- **Quadruple therapy**

 PPI
 Bismuth subcitrate
 Tetracycline
 metronidazole

During your 3rd year clinical rotation, you are introduced to JR, a 56-year-old oil executive who has been a chain smoker (1/2 pack per day for 20 years), and for the past few years has been experiencing a mid-epigastric upper abdominal pain that is diminished by eating or taking antacid tablets. JR drinks three or four glasses of single malt Scotch every evening, as well as an occasional two martini lunch. JR has been taking 10-20 full strength aspirin tablets per week for stress-related headaches. He is unaware of any signs of blood in the stool. JR is advised to replace his aspirin with acetaminophen, reduce his drinking and smoking, and is advised to start taking **famotidine 40 mg bid**. In terms of treating his peptic ulcer disease, what beneficial goal would you hope to achieve with this therapy?

a. Eradication of H. pylori
b. Neutralize gastric acid
c. Protect the gastroduodenal mucosa
d. Reduce the amount of acid secretion

reduce the amount of acid secretion

A patient with a history of dyspepsia has a positive urea breath test and a positive stool antigen test for the presence of H. pylori. While multiple treatment regimens have been evaluated for this type of infection, the most common regimen being used in areas where antibiotic resistance is low is "triple therapy" consisting of a proton pump inhibitor (PPI), amoxicillin and an additional drug. Which drug is the most common 3rd component used for triple therapy?

a. Cimetidine
b. Clarithromycin
c. Loperamide
d. Metronidazole
e. Pirenzepine

clarithromycin

You are considering treatment for a patient recently diagnosed with stomach ulcer caused by H. pylori infection. The patient has a history of anaphylactic reactions to penicillins. What drug would you use as a substitute for amoxicillin in "triple therapy" that exerts its affect by directly targeting bacterial DNA?

a. Doxycycline
b. Metronidazole
c. Quinupristin
d. Vancomycin

metronidazole

A 37-year-old man who was started on methadone therapy for opioid addiction a week ago is brought to the Emergency Department suffering from excessive drowsiness and disorientation. During the physical exam you notice that his breasts appear abnormally enlarged. He insists that he has not been taking more methadone than prescribed but admits that he has been taking an over-the-counter H2 blocker daily to prevent "stomach problems" for the past year. You suspect a drug interaction may explain his current condition. Which drug is most likely responsible?

a. Cimetidine
b. Famotidine
c. Nizatidine
d. Rantitidine

cimetidine

A 65-year-old woman with a history of rheumatoid arthritis visits your clinic complaining of increasing stomach pain that started several weeks ago. She recently began taking a half dozen or more 400 mg ibuprofen tablets daily to help control her progressively increasing arthritic symptoms. Her lab results indicate a blood hemoglobin of 7 g/dl (normal 12-15 gm/dl), hematocrit 29% (normal 38-46%), as well as a guaiac-positive stool test. You discuss treatment with a synthetic prostaglandin that would be contraindicated if she were much younger & not on contraceptives. Which drug is this?

a. Dronabinol
b. Misoprostol
c. Omeprazole
d. Ondansetron
e. Sulcralfate

misoprostol

A 42-year-old man has been experiencing heartburn like pain after meals, especially when reclining on his sofa after a meal. His symptoms are relieved by antacids. You discuss therapeutic options, which include either a H2 receptor antagonist or a proton pump inhibitor as a drug of first choice. You also discuss the use of an additional agent that can provide a protective coating over necrotic tissue, which can help with healing of ulcers. What agent has this mechanism of action?

a. Calcium Carbonate
b. Ranitidine
c. Loperamide
d. Pirenzapine
e. Sucralfate

sucralfate

Four weeks into therapy for peptic ulcer, the patient complained of difficulty in erection during sex. He also noticed a bilateral enlargement of his breast. Which of the following drugs was most likely prescribed?

a. Famotidine
b. Omeprazole
c. Sucralfate
d. Cimetidine
e. Bismuth
f. Misoprostol

Cimetidine

THANK YOU